

MATHS



Topic- PERCENTAGE प्रतिशत

Lecture No.- 01

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Recap of Previous Lecture



Topic

INTRO

Topic

Topic

Topic

Topic

Topics to be Covered



Topic



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PERCENTAGE (प्रतिशत)



PERCENTAGE



PERCENTAGE (प्रतिशत)



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PERCENTAGE (प्रतिशत)



TYPE-7

EXAM PASS FAILBASED QUESTIONS

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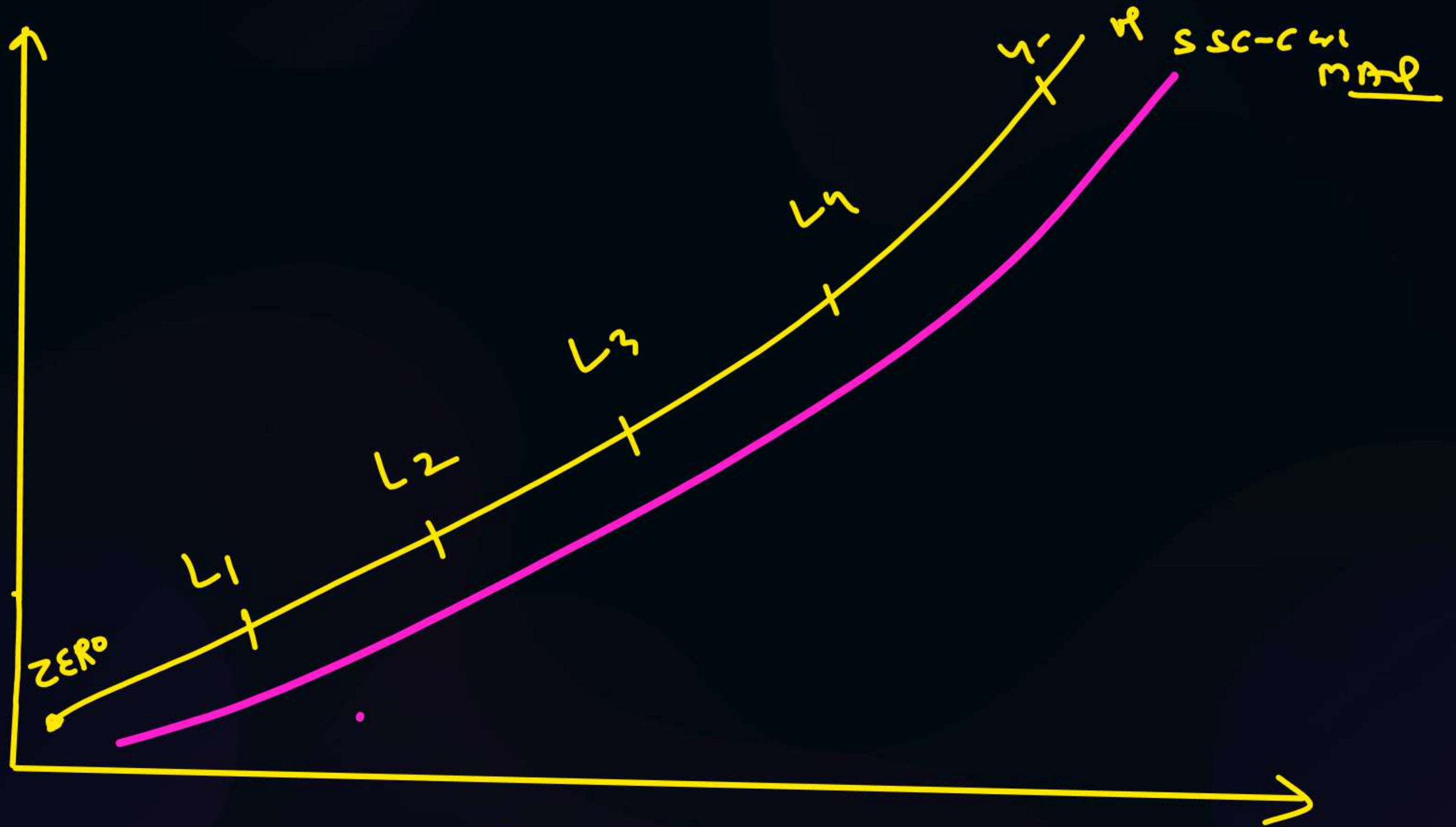
TYPE-11

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PERCENTAGE (प्रतिशत)

FRACTION	PERCENTAGE	DECIMAL
1	100%	100%
$\frac{1}{2}$	$\longrightarrow 50\%$	
$\frac{1}{3}$	$\longrightarrow \underline{33\frac{1}{3}\%} = \underline{33.33\%}$	
$\frac{1}{4}$		
$\frac{1}{5}$		
$\frac{1}{6}$		
$\frac{1}{7}$		
$\frac{1}{8}$		
$\frac{1}{9}$		
$\frac{1}{10}$		



PERCENTAGE (प्रतिशत)

FRACTION	PERCENTAGE	DECIMAL
1	100%	100%
$\frac{1}{2}$	50%	50%
$\frac{1}{3}$	$33\frac{1}{3}\%$ = $33 + 0.333 = 33.33\%$	33.33%
$\frac{1}{4}$	25%	25%
$\frac{1}{5}$	20%	20%
$\frac{1}{6}$	$16\frac{2}{3}\%$	16.66%
$\frac{1}{7}$	$14\frac{2}{7}\%$	14.28%
$\frac{1}{8}$	$12\frac{1}{2}\%$	12.5%
$\frac{1}{9}$	$11\frac{1}{9}\%$	11.11%
$\frac{1}{10}$	10%	10%

$$\frac{1}{3} = 0.333$$

$$\begin{aligned}16\frac{2}{3}\% &= 16 + \frac{2}{3} \\ &= 16 + 0.66 \\ &= 16.66\%.\end{aligned}$$



PERCENTAGE (प्रतिशत)

FRACTION	PERCENTAGE	DECIMAL
$\frac{1}{11}$	$9\frac{1}{11}\%$	9.09%
$\frac{1}{12}$	$8\frac{1}{3}\%$	8.33%
$\frac{1}{13}$	$7\frac{9}{13}\%$	7.69%
$\frac{1}{14}$	$7\frac{1}{7}\%$	7.14%
$\frac{1}{15}$	$6\frac{2}{3}\%$	6.66%
$\frac{1}{16}$	$6\frac{1}{4}\%$	6.25%
$\frac{1}{17}$	$5\frac{15}{17}\%$	5.88%
$\frac{1}{18}$	$5\frac{5}{9}\%$	5.55%
$\frac{1}{19}$	$5\frac{5}{19}\%$	5.26%
$\frac{1}{20}$	05%	0.05%



PERCENTAGE (प्रतिशत)



SOME DIFFERENT VERY USEFUL FRACTION





PERCENTAGE (प्रतिशत)

FRACTION	PERCENTAGE	DECIMAL
$\frac{1}{24}$	$4\frac{1}{6}\%$	4.16%
$\frac{3}{8}$	$37\frac{1}{2}\%$	37.5%
$\frac{5}{8}$	$62\frac{1}{2}\%$	62.5%
$\frac{4}{7}$	$57\frac{1}{7}\%$	57.14%
$\frac{5}{7}$	$71\frac{3}{7}\%$	71.42%
$\frac{2}{3}$	$66\frac{2}{3}\%$	66.66%



PERCENTAGE (प्रतिशत)

PART 1 FRACTION TO % CONVERSION



PERCENTAGE (प्रतिशत)



$$= \frac{1 \times 100}{2} = \% ?$$
$$= \underline{50\%}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{1}{5} = 20\%$$

$$\frac{4}{5}$$

$$\frac{4 \times 20}{5} = 80\%$$

$$\begin{aligned} &= 4 \times \left(\frac{1}{5} \right) \\ &= 4 \times 20\% \\ &= 80\% \end{aligned}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\begin{aligned} \frac{3}{4} &= 3 \times \left(\frac{1}{4}\right) \\ &= 3 \times 25\% \\ &= \underline{75\%} \end{aligned}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{1}{9} = 11\frac{1}{9}\%$$
$$= 11.11\%$$

$$\frac{4}{9} = 4 \times \frac{1}{9}$$
$$= 4 \left[11\frac{1}{9}\% \right]$$
$$= 44\frac{4}{9}\%$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\begin{aligned} 7 \times \frac{1}{9} &= 7 \times \frac{1}{9} \\ 7(11.11\%) &= 7 \left[11 \frac{1}{9} \% \right] \\ \underline{77.77\%} &= 77 \frac{7}{9} \% \end{aligned}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{5}{2} \\ 2\frac{1}{2}$$

$$\frac{5}{8}$$

$$= 5 \times \frac{1}{8}$$

$$= 5 \left[12\frac{1}{2} \cdot \frac{1}{1} \right]$$

$$= 60 \frac{5}{2} \%$$

$$= 60 + 2\frac{1}{2}$$

$$= 62\frac{1}{2} = 62.5\%$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{5}{2}$$
$$2\frac{1}{2}$$

$$\frac{5}{8}$$

$$= 5 \times \frac{1}{8}$$

$$= 5 \left[12\frac{1}{2} \% \right]$$

$$= 60\frac{5}{2} \% = 60 + 2\frac{1}{2}$$

$$= 62\frac{1}{2} \%$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{5}{6} = 5 \times \frac{1}{6}$$

$$= 5 \left[16 \frac{2}{3} \% \right]$$

$$= 80 \frac{10}{3} \% = 80 + 3 \frac{1}{3}$$

$$= \underline{83 \frac{1}{3} \%}$$

$$\frac{1}{6} = 16 \frac{2}{3} \%$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{1}{15} = 6\frac{2}{3}\%$$

$$\frac{7}{15}$$

$$= 7 \times \frac{1}{15}$$

$$= 7 \left[6\frac{2}{3}\% \right]$$

$$= 42 + \frac{14}{3}\% = 42 + 4\frac{2}{3}\%$$

$$= 46\frac{2}{3}\%$$



PERCENTAGE (प्रतिशत)

Q. CONVERT FRACTION TO PERCENTAGE ?

$$\frac{11}{12}$$

$$\begin{aligned} &= 1 - \frac{1}{12} \\ &= 100\% - \left(8\frac{1}{3}\right)\% \\ &= 91\frac{2}{3}\% \end{aligned}$$

$$\begin{aligned} &= 11 \times \frac{1}{12} \\ &= 11 \left[8\frac{1}{3} \cdot 1 \right] \\ &= 88 + \frac{11}{3} \\ &= 88 + 3\frac{2}{3} \\ &= 91\frac{2}{3}\% = 91.66\% \end{aligned}$$

$$= 10\% - 2\%$$

$$= 8\%$$

$$= 10 - \left(8 \frac{2}{3} \right)$$

$$= 1 \frac{1}{3}$$

$$10 - 7$$

$$3$$

$$= 20 - \left(5 \frac{4}{19} \right)$$

$$= 14 \frac{15}{19}$$

$$8$$



PERCENTAGE (प्रतिशत)



PART 2 % TO FRACTION CONVERSION



PERCENTAGE (प्रतिशत)

Q. CONVERT % TO FRACTION ?

$$100\% = 1$$

$$6\frac{2}{3}\% = \frac{1}{15}$$

$$\begin{aligned} 106\frac{2}{3}\% &= 100\% + 6\frac{2}{3}\% \\ &= 1 + \frac{1}{15} \\ &= \frac{16}{15} \text{ or } 1\frac{1}{15} \end{aligned}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT % TO FRACTION ?

$$100\% = 1$$

$$300\% = 3$$

$$366\frac{2}{3}\%$$

$$= 300\% + 66\frac{2}{3}\%$$

$$= 3 + \frac{2}{3} = \frac{11}{3}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT % TO FRACTION ?

$$616\frac{2}{3}\% = 600\% + 16\frac{2}{3}\%$$
$$= 6 + \frac{1}{6}$$
$$= \frac{37}{6}$$

le



PERCENTAGE (प्रतिशत)

Q. CONVERT % TO FRACTION ?

$$906\frac{1}{4}\%$$

$$= 900\% + 6\frac{1}{4}\%$$

$$= 9 + \frac{1}{16} = 9\frac{1}{16}$$



PERCENTAGE (प्रतिशत)

Q. CONVERT % TO FRACTION ?

$$11\frac{1}{9} = 11.11\%$$

$$811.11\%$$

$$= 800\% + 11.11\%$$

$$= 8 + \frac{1}{9}$$

$$\therefore 8\frac{1}{9} = \frac{73}{9}$$

$$= 16\frac{2}{3}\%$$

$$= \frac{52 \times 1}{3 \times 100}$$

$$= \frac{1}{6}$$



PERCENTAGE (प्रतिशत)

Q. FIND THE VALUE OF ?

$$\underline{37\frac{1}{2}}\% \text{ of } 800 =$$

$$\frac{3}{2} \times \frac{100}{\cancel{100}} = \underline{300}$$



PERCENTAGE (प्रतिशत)

Q. FIND THE VALUE OF ?

$$57\frac{1}{7}\% \text{ of } 777 =$$

$$\frac{4}{7} \times 777 = \underline{444}$$



PERCENTAGE (प्रतिशत)



PART 3 MULTIPLE OF FRACTION CONVERSION



PERCENTAGE (प्रतिशत)

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1	100%	100%
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$\frac{1}{4}$	25%	25%
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$\frac{1}{7}$	$14\frac{2}{7}\%$	14.28%
$\frac{1}{8}$	$12\frac{1}{2}\%$	12.5%
$\frac{1}{9}$	$11\frac{1}{9}\%$	11.11%
$\frac{1}{10}$	10%	10%

$$\frac{2}{7}$$

$$= 28.56\%$$

$$= 2 \times 14.28$$

$$= 2 \times \frac{1}{7}$$

$$= \frac{2}{7}$$



PERCENTAGE (प्रतिशत)

FRACTION	PERCENTAGE	DECIMAL
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$\frac{1}{14}$	$7\frac{1}{7}\%$	7.14%
$\frac{1}{15}$	$6\frac{2}{3}\%$	6.66%
$\frac{1}{16}$	$6\frac{1}{4}\%$	6.25%
$\frac{1}{17}$	$5\frac{15}{17}\%$	5.88%
$\frac{1}{18}$	$5\frac{5}{9}\%$	5.55%
$\frac{1}{19}$	$5\frac{5}{19}\%$	5.26%
$\frac{1}{20}$	05%	0.05%



PERCENTAGE (प्रतिशत)

Q. CONVERT % TO FRACTION ?

42.84 % of 6300

SSC 2020

$$\begin{aligned} &= 3 \times 14.28 \% \times 6300 \\ &= 3 \times \frac{1}{7} \times \cancel{6300}^{700} \\ &= \underline{\underline{2700}} \end{aligned}$$



PERCENTAGE (प्रतिशत)

$$45.45 \% \text{ of } 121 =$$

$$\begin{aligned} &= 5 \left[9.09\% \right] \times (121) \\ &= 5 \times \left(\frac{1}{11} \right) \times 121 \\ &= 55 \\ &\quad \checkmark \checkmark \end{aligned}$$



PERCENTAGE (प्रतिशत)

$$36.36 \% \text{ of } 2200 =$$

$$\begin{aligned} \frac{36.36}{4} &= 4 [9.09\%] \times 2200 \\ 9.09\% &= 4 \times \frac{1}{11} \times 2200 = 800 \end{aligned}$$

$$\begin{aligned} &= 36.36\% \\ &= 4 [9.09] \\ 4 \sqrt{36.36} & (9.09) \\ \underline{36} & \\ & \times 36 \\ & \underline{36} \end{aligned}$$



PERCENTAGE (प्रतिशत)



BASIC % BASED QUESTIONS



PERCENTAGE (प्रतिशत)

Q. If 40% of $(A + B) = 60\%$ of $(A - B)$, Find $\frac{2A-3B}{A+B}$

यदि $(A + B)$ का 40% = $(A - B)$ का 60% है, तो $\frac{2A-3B}{A+B}$ ज्ञात कीजिए

$$(A+B) \frac{40}{100} = (A-B) \frac{60}{100}$$

$$2A + 2B = 3A - 3B$$

$$\frac{5B}{1} = \frac{A}{1}$$

$$= \frac{2 \times 5 - 3 \times 1}{5+1}$$

$$= \frac{7}{6}$$

(a) 6/7

(b) 6/5

(c) 5/6

(d) 7/6

D



PERCENTAGE (प्रतिशत)

Q. 5% of $x = y$, then $y\%$ of 20 is same as?

x का 5% = y , तो 20 का $y\%$ क्या है? CGL MAINS

(A) $\frac{20}{100} \times \frac{x}{2}$

$\frac{x}{10}$

$$\frac{x \times 5}{100} = y$$

$$\frac{x}{20} = \frac{y}{5}$$

$$\frac{x}{100}$$

$$\frac{20 \times y}{100 \times 5}$$

$$\frac{y}{5}$$

(D)

$$\frac{20}{100} \times \frac{x}{2} = \frac{x}{10}$$

- (a) 20% of $x/2$
- (b) 50% of $x/20$
- (c) 50% of $x/2$
- (d) 20% of $x/20$



PERCENTAGE (प्रतिशत)



Q. 10% of $x = y$, then $y\%$ of 40 is same as?

x का 10% = y , तो 40 का $y\%$ किसके समान है?

$$\frac{x \times 10}{100} = y$$

$$= \frac{40 \times y}{100 \times 100}$$

$$= \frac{4x}{100}$$

(D)

$$\frac{40 \times y}{100 \times 100} = \frac{4x}{100}$$

(a) 60% of $x/10$

(b) 50% of $x/10$

(c) 30% of $x/10$

(d) 40% of $x/10$



2 mins Summary



Topic

Topic

Topic

Topic

Topic

PER-TABLE | Bop
+
/

THANK - YOU

A short, horizontal yellow line.